

Chapter 136

Color Warper

The Color Warper is a powerful toolset for making both highly specific adjustments to particular things in the frame, and widely general adjustments to create unique looks.

The two modes of the Color Warper each make it easy to change hues using the Chroma Warp, or simultaneously modify two different color attributes, either saturation and hue, or lightness and hue. This gives the Color Warper an advantage over curves, which only let you adjust one color characteristic at a time.

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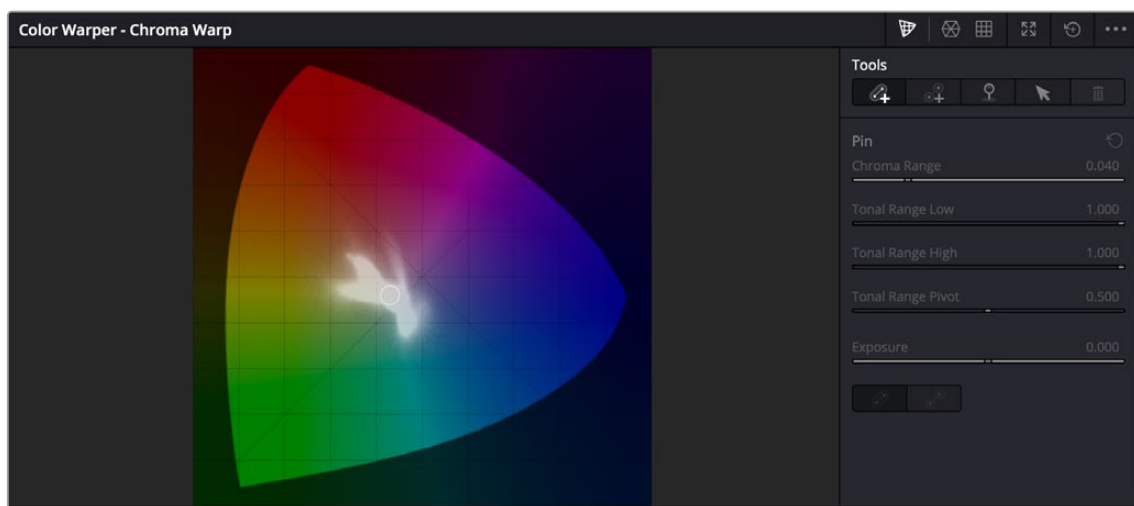
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Introduction to the Color Warper

The Color Warper consists of two modes: Chroma Warp, and Hue-Saturation, Chroma-Luma. Both modes are designed to warp one set of colors into another with a smooth falloff that might be difficult to achieve with qualifiers alone. However, for each node, you must choose between using the Chroma Warp, or the Hue-Saturation and Chroma-Luma controls.

Chroma Warp

The Chroma Warp is a mode found in the Color Warper palette. Like the Color Warper hue-sat/chroma-luma controls, its purpose is to make smooth and precise changes between color ranges. The main difference is that the Chroma Warp uses strokes on a chromaticity diagram rather than adjusting the points on a mesh to make its changes. Both will give you great results, but you can only use one of these tools per node, either the traditional Color Warper or the Chroma Warp.



The Chroma Warp chromaticity diagram on the left and toolset to the right

The main interface of the Chroma Warp contains a chromaticity diagram with the range of the current colors in the image overlaid on top in white (similar to a vectorscope representation), with the toolset to the left. The chromaticity diagram is perceptually uniform, meaning that the distances between points on the diagram closely align to the visual differences seen in the image when using the Chroma Warp. The Chroma Warp is exposure independent, meaning once you've set a Chroma Warp, subsequent changes in luminance will not break your qualified selection.

The first decision to make is whether you want to use the Chroma Warp in Normal or Point to Point modes.

Using The Chroma Warp in Normal Mode

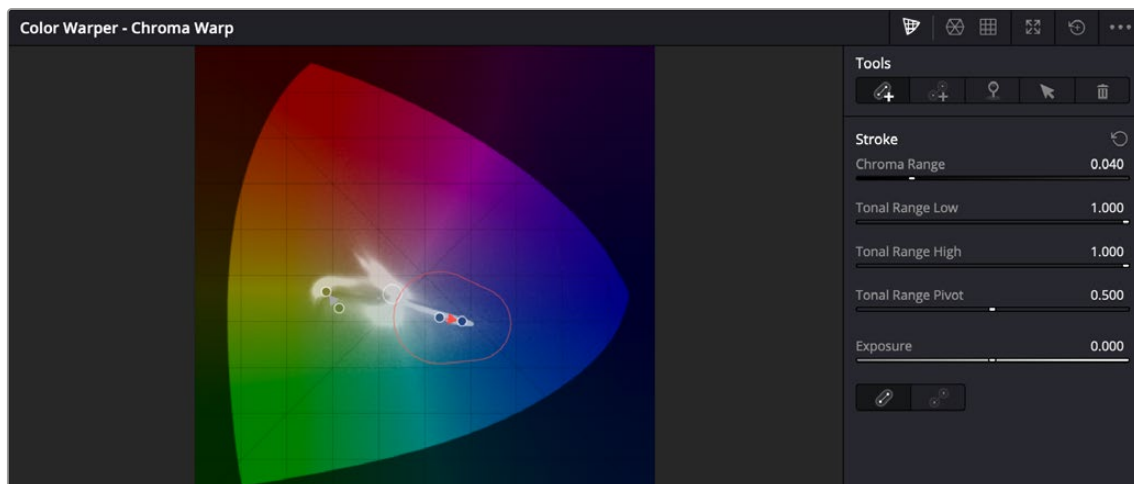
Use this mode when you want to naturally transition from one color to another through the entire color range that lies between them, for example changing the color of an entire sky to a deeper blue.

Select the color range you want to modify by:

- Clicking the original color using the qualifier eyedropper in the Viewer, then dragging towards the direction of the new color in the chromaticity diagram. The Viewer will update as you drag through the color ranges, allowing you to easily dial in the exact change you want.
- Clicking on the color representation inside the chromaticity diagram itself, then dragging towards the color you wish to change it to.

In either case, the initial color selected will be represented by a blue dot on the chromaticity diagram, and the final color will be represented by an orange dot. They will be connected by an arrow showing the range of change. An orange outline around the vector shows the entire range of colors modified.

Dragging the points toward the white point (the small circle in the diagram) will desaturate the color, while dragging it toward the edges will increase a color's saturation.



The Chroma Warp in Normal mode, changing both the tonal range of the grass to more yellow and the sky range to deeper blue.

Using The Chroma Warp in Point to Point Mode

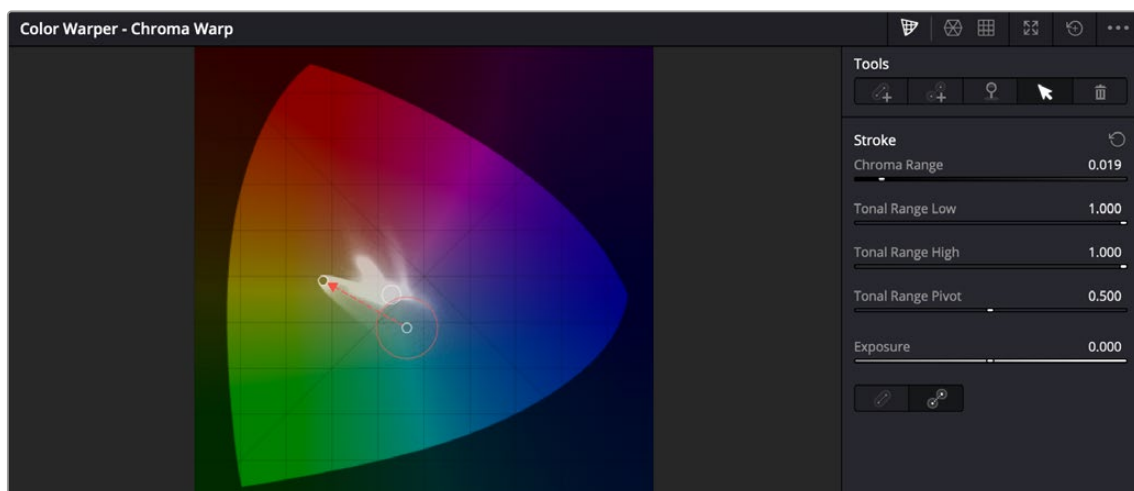
Use this mode to isolate and change a specific color to another color entirely, for example changing the color of a shirt from blue to yellow without going through green.

Select the specific color you want to modify by:

- Clicking the original color using the qualifier eyedropper in the Viewer, then dragging towards the direction of the new color in the chromaticity diagram. The Viewer will update as you drag through the colors, allowing you to easily dial in the exact destination color you want.
- Clicking on the color representation inside the chromaticity diagram itself, then dragging towards the color you wish to change it to.

In either case, the initial color selected will be represented by a blue dot on the chromaticity diagram, and the final color will be represented by an orange dot. They will be connected by an arrow showing the range of change. An orange outline around both dots shows the selection range of the original and final colors.

Dragging the points toward the white point (the small circle in the diagram) will desaturate the color, while dragging it toward the edges will increase a color's saturation.



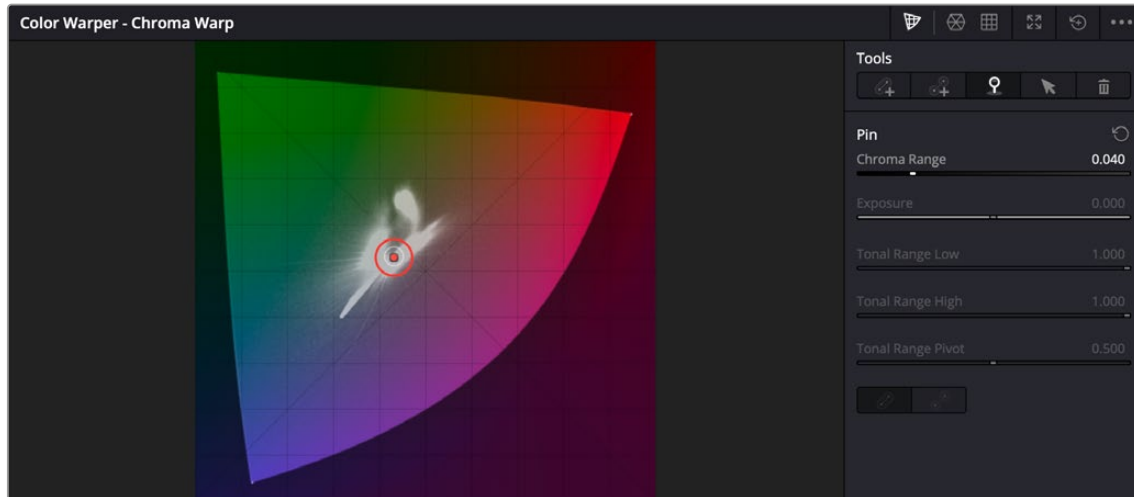
The Chroma Warp in Point to Point mode, changing just the color of the shirt from blue to yellow.

TIP: You can use multiple Normal and Point to Point selections concurrently for very advanced color changes within the same node.

Pin Points

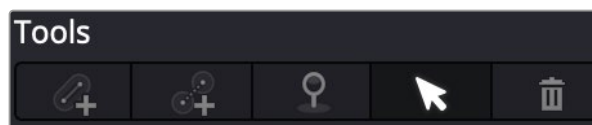
This tool allows you to set a pin designating a color range to be excluded from the selected range. For example, if you wanted to change the color of a sign from red to green, but it had yellow lettering that fell into the range between, you could add a pin point to the yellow text to exclude it from the color change. Another use case would be to put a pin point on the skin tones of an actor so that their skin remained the same regardless of the color changes happening around it.

You can set a pin point just like the selection above, either by using the qualifier on a color in the image itself in the Viewer, or by clicking on the color in the chromaticity diagram.



Placing a Pin Point around the white point to keep neutral tones from tinting during a Chroma Warp operation

TIP: If using the tool is causing tinting issues, try placing a Pin Point on the white point of the chromaticity diagram (the center circle). This will remove all neutral grays from the color adjustment.



The Chroma Warp tools

Chroma Warp Tools

The Chroma Warp toolset are where you access the main color selection parameters of the Warp.

- **Add Stroke (Normal):** This puts the Chroma Warp in Normal mode to select a color range.
- **Add Stroke (Point to Point):** This puts the Chroma Warp in Point to Point mode to select a specific color.
- **Add Pin Point:** This allows you to set a pin designating a color to be excluded from the selected range.
- **Select:** Lets you click on an existing range and drag it around the chromaticity diagram.
- **Delete Strokes or Points:** This lets you delete specific strokes and points you have already set on the Chroma Warp. Simply click on one of the points of a range, then click on the trashcan icon.

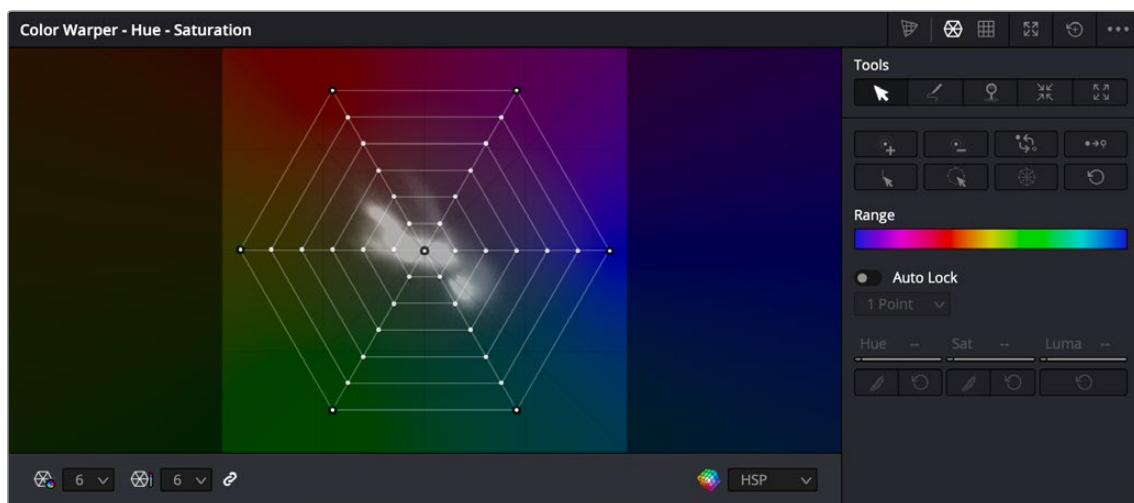
Chroma Warp Stroke

The Chroma Warp stroke tools modify and refine any selection you've made.

- **Chroma Range:** Adjust this slider to set the amount of adjacent colors included in the range. Lower ranges make the color selection more specific, while higher ranges encompass more of the adjacent colors.
- **Exposure:** Adjusting this slider darkens or brightens the selection in the ending circle only (the orange dot). Negative values darken, while positive values brighten.
- **Tonal Range Low:** Adjusting this slider lets you primarily affect the darker regions of the initially selected color for change, leaving the brighter regions unchanged. This adjustment is global and applies to all strokes in the chromaticity diagram.
- **Tonal Range High:** Adjusting this slider lets you primarily affect the brighter regions of the initially selected color for change, leaving the darker regions unchanged. This adjustment is global and applies to all strokes in the chromaticity diagram.
- **Tonal Range Pivot:** This slider lets you set the initial point between Tonal Range Low and Tonal Range High that those tools diverge from.
- **Normal/Point to Point Toggle:** Clicking on either of these icons changes a Normal selection to a Point to Point or vice versa.

Hue-Saturation/Chroma-Luma Mode

The Color Warper Hue-Saturation and Chroma-Luma palettes are a mesh-based warping tool that, instead of warping the spatial location of pixels, warps one set of colors into another. Better yet, unlike the HSL curves, they let you simultaneously adjust two color parameters at the same time, within user-selectable regions of specific hues. These adjustments, made using a grid of draggable control points, automatically have a smooth falloff from the colors you're warping to other colors that are locked into place. The smoothness of this falloff depends on the distance between the warp points that are being adjusted and other warp points that are locked in place to prevent change.



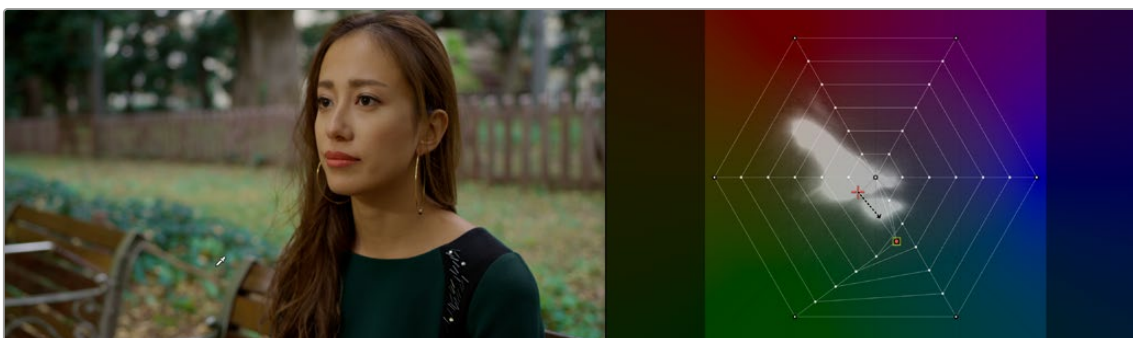
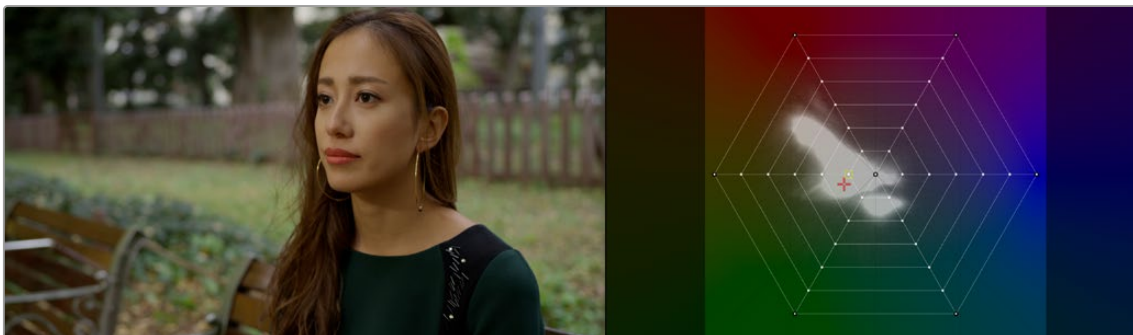
The Color Warper, located in the center palette area

There are two ways you can use this tool. You can drag the control points in the grid to make highly specific image adjustments. More intuitively, you can click to sample the image, which selects the closest grid point that affects that color, and then drag to make an adjustment with that control point.



Clicking and dragging to sample skin tone in the Viewer to adjust that color's control point on the Hue-Saturation warping grid

You can use this tool to make extremely specific adjustments to colors in an image, similar to what you might do with a Qualifier key or with the HSL curves. However, by warping the points using a grid, you can manipulate two color characteristics at the same time, adjusting both hue and saturation, or chroma and luma. In the following example, specific adjustments are being made to refine the woman's shirt, to adjust her skin tone, and to manipulate the greens of the foliage in the shot. All three of these naturalistic adjustments are smooth and artifact-free by virtue of the Color Warper's method of cleanly remapping color.



(Top) The original image, (Bottom) Manipulating the hue and saturation of the grass using the Chroma-Luma controls

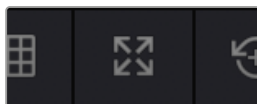
You can also use this tool to create wide-ranging stylistic looks for the overall image, bending the color in creative ways that other controls don't. In the following image, the less saturated and neutral colors of the image are all warped towards a bluish-cyan, while the more saturated oranges are warped towards a saturated orange-red, to quickly create a stylistic treatment of the image.



(Top) The original image, (Bottom) Stylizing the overall image using the Hue-Saturation controls

The Color Warper Interface

You can either use the Color Warper as it appears docked into the middle palette area, or you can click the Expand button to open the Color Warper into a floating window that you can make as large as you like, giving you more detailed control of dense grids of control points. For convenience, the examples in this chapter show the Color Warper side-by-side with the Viewer, in a floating window. While the Color Warper is open in its own window, clicking the Close button puts the palette back in the palette area below.

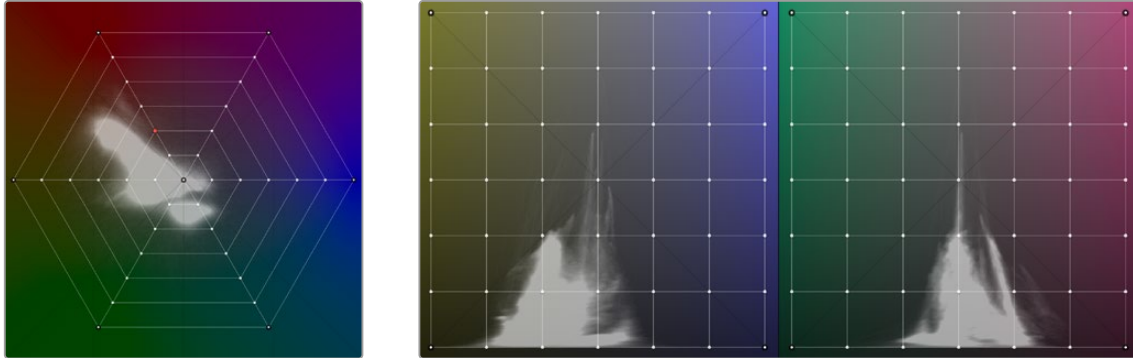


Clicking the Expand button opens the Color Warper into a floating window, clicking the Close button in the floating window puts the palette back.

There are two overall regions of the Color Warper palette. At the left, the grid area has the actual warping grid with points you manipulate to make adjustments, while controls underneath let you choose the resolution of the grid and the color space in which it works. At the right, the control area provides all the tools, range controls, and feathering controls that are available for manipulating the warping grid in a variety of highly specific ways.

Using the Grid Controls to Manipulate Color

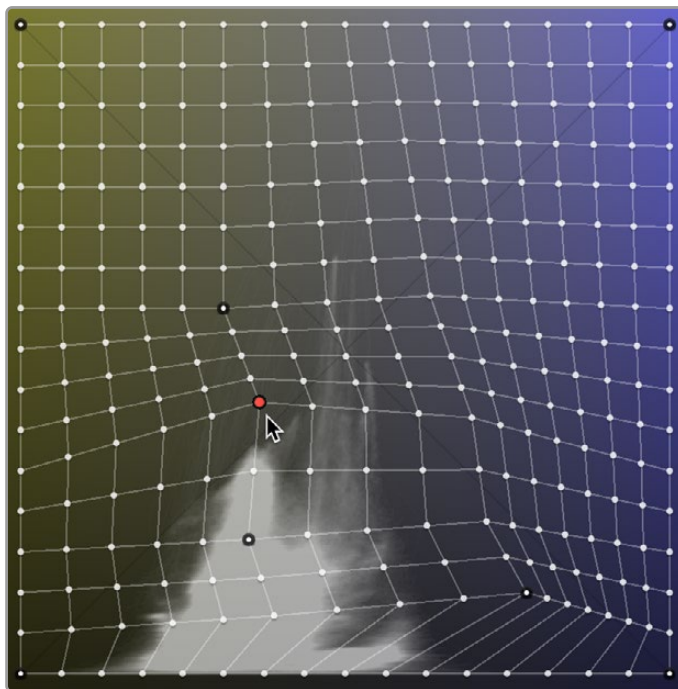
This palette has two modes; each lets you manipulate the colors of the image in different ways, and each has a different style of grid used for warping the different color channels that are affected. Learning how to manipulate the color warping points of each of these grids is central to understanding how this palette works.



(Left) The radial grid in Hue-Saturation mode,
(Right) The two rectangular grids in Chroma-Luma mode

Dragging to Warp Colors

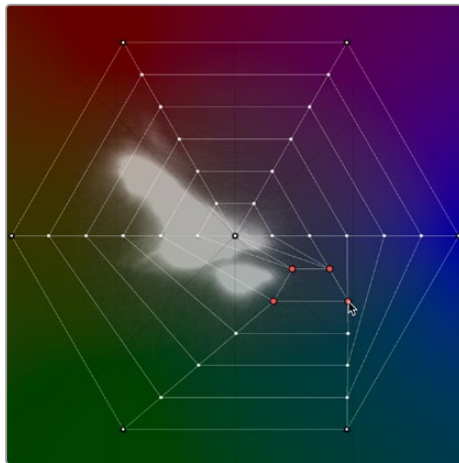
Despite the different shapes, each of these grids are manipulated in similar ways. Dragging a control point alters whatever color corresponds to that point, while the colors that surround this dragged point are also adjusted proportionally according to their location between the point being dragged and surrounding points that either (a) aren't affected, or (b) are locked into place. Locked points prevent the colors at those points from being changed. By locking some parts of a grid and dragging other parts, you can create sophisticated hue-specific adjustments.



The orange control point has been selected and dragged to make an adjustment. Control points with black outlines are locked in place and help prevent control points on the opposite side from being moved. All other control points between the selected and locked control points are stretched or squished depending on the selected point's adjustment.

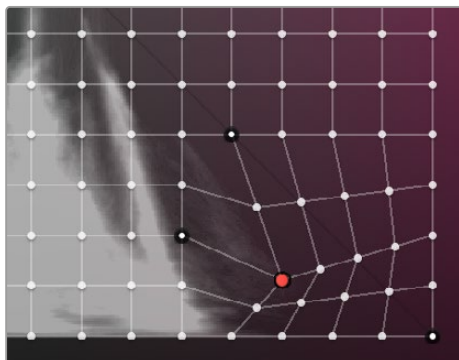
The image above shows the three states of control points in the grid:

- **Orange:** *Selected* points appear orange. You can select multiple points by Command-clicking multiple points, by dragging a bounding box over multiple points, or by using the draw selection tool to click and draw over any points you want to select. Once you've selected and moved a control point, it becomes locked to keep that adjustment in place.



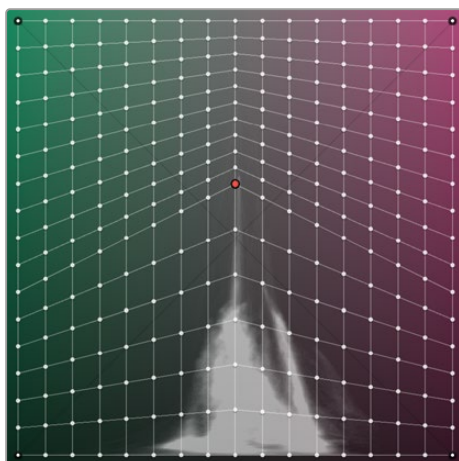
Selected points appear orange; in this example four points are selected to move them all together.

- **Outlined:** White control points with a black outline are *locked*. Locked points do not move automatically when nearby points are dragged, so locked points are important for preventing specific colors you don't want altered from being changed, as well as for saving changes to points that you've adjusted. All adjusted points become locked. The outer corners of the Chroma-Luma grid are locked, as is the outer ring and center point of the Hue-Saturation web.



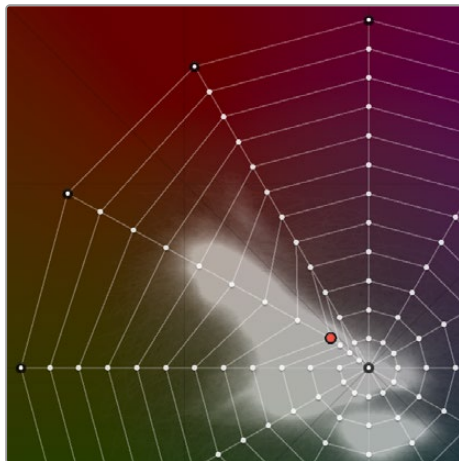
Locked points with black outlines keeping other colors in the image from being adjusted by the selected point

- **White:** White control points are neither selected nor locked, and will be stretched or squished when adjacent control points are moved.



Unlocked control points between a selected point and locked points (in this case the corners of the grid) will squish and stretch to warp an entire range of color as you drag.

While control points work much the same in either the Hue-Saturation or Chroma-Luma modes, there are some important differences. The radial “spokes” of the Hue-Saturation wheel don’t influence neighboring spokes, so any adjustment you make to a control point on one spoke only affects the colors that fall between it and the two neighboring spokes that surround it. This means it’s not necessary to lock control points on neighboring spokes of this wheel-shaped grid to prevent them from changing.



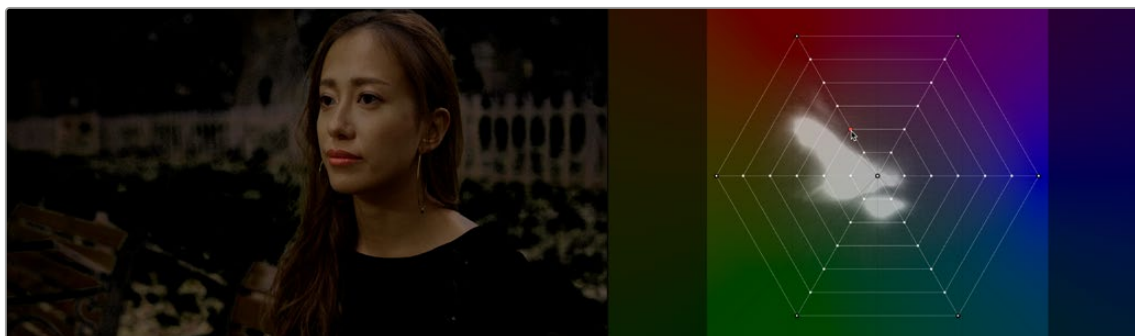
Adjustments made to control points on one “spoke” of the Hue-Saturation wheel grid don’t affect those on neighboring spokes, which act as if they’re locked, even though they’re not.

Both the Hue-Saturation and Chroma-Luma warper grids lock the outer boundary of the grid. You can still drag these points to adjust them, but they can’t be unlocked and they won’t move automatically when you adjust other control points. The Hue-Saturation grid has an additional point locked right in the center, which keeps the neutral colors (black through gray and white) neutral.

TIP: Dragging the locked center point of the Hue-Saturation grid lets you alter the color of the white point, while proportionally warping all other colors in the image to smoothly accommodate this change. This can be the starting point of a lot of different looks.

Previewing Which Colors Are Warped By Each Control Point

If you press and hold the Option key while you click on a control point in the warp grid, you get a preview in the Viewer of which colors that control point will affect. Affected pixels appear in color against black, which represents all pixels that will remain unaffected by that control point.

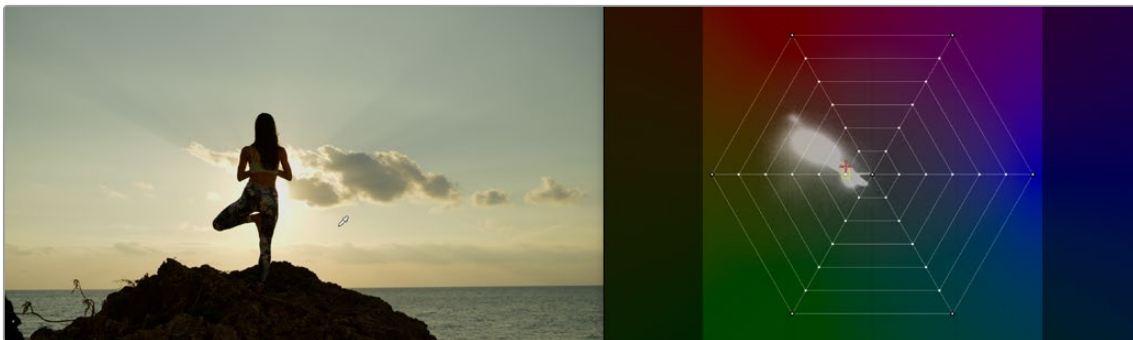


Holding the Option key down while clicking a Color Warper control point shows a preview of affected pixels against black.

Sampling to Warp Colors

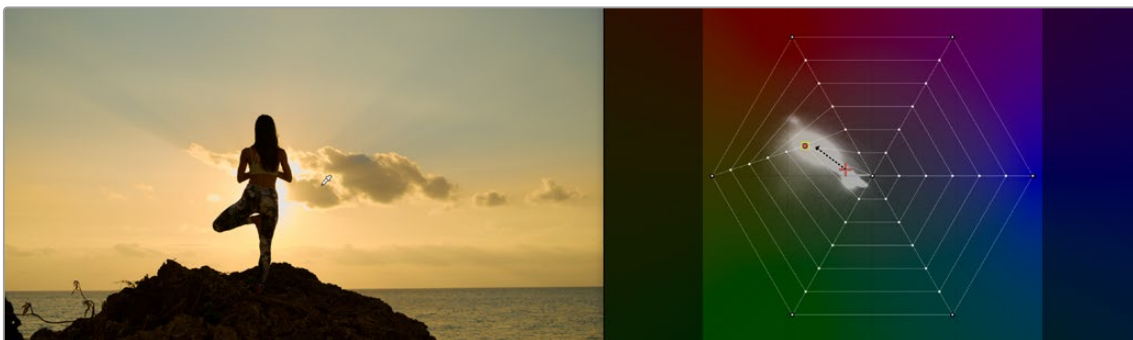
While it's important to learn how control points in the grid are manipulated, the most intuitive way of working with this control is to use the pointer to click within the Viewer to sample a color you want to adjust, hold the pointer button down, and then drag to adjust the color.

When moving the pointer in the Viewer, you'll see a crosshairs moving around the warping grid that shows you the exact color on the vectorscope graph, while a yellow box indicates the control point that's closest to the color you're sampling that will be selected if you click. In the following example, positioning the eyedropper over the sky shows that clicking will select a control point close to the center of the grid.



Moving the pointer in the Viewer shows a preview in the warping grid of which control point you'll select were you to click.

Clicking and dragging selects and then moves the control point corresponding to that color in the Viewer. As you drag the pointer, you'll see the selected control point move in the same direction as the mouse while the colors of the image update in real time to show the adjustment you're making. Meanwhile, an arrow in the grid shows you the delta of the current adjustment. Moving a selected control point to another hue in the grid warps the original color to the new color.



Clicking and dragging moves the control point corresponding to that color in the grid to adjust the color and locks it into place.

You can also lock control points by sampling the image. Hold the Command and Shift keys down, and then click on the color or colors you want to lock in the warp grid to prevent them from being changed. This makes it easy to work directly in the Viewer to lock colors you don't want to change, before adjusting other colors that you do want to change.

In the following example, Command-Shift-clicking on the woman's shoulder locks the yellowish highlights falling on her uniform in place. Then, clicking and dragging a highlight on her face lets us push the highlights towards teal without losing the warmer highlights on her shoulder, all by sampling directly in the Viewer.



(Top) Command-Shift-clicking to lock the closest control point to a color on the warping grid you don't want to change, (Bottom) Warping a neighboring color while the control point you just locked keeps that part of the image from changing

Resetting Grid Points

If, at any point, you adjust a point that you want to reset, simply right-click on that point to de-lock and reset it.

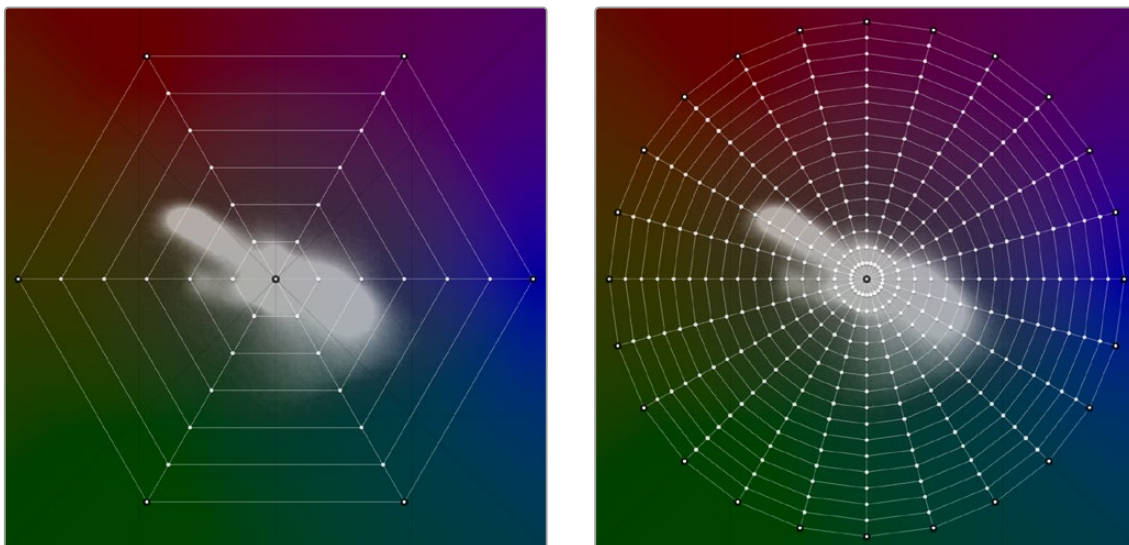
Grid Resolution Affects The Specificity and Quality of Adjustments

You can change the resolution of the grids in either Hue-Saturation or Chroma-Luma modes, using the controls at the bottom of the Color Warper palette, underneath the grid itself. Separate drop-down menus let you choose the resolution of control points for hue and saturation, or for chroma and luma separately, although by default they're linked together.



The two resolution controls found underneath the grid controls

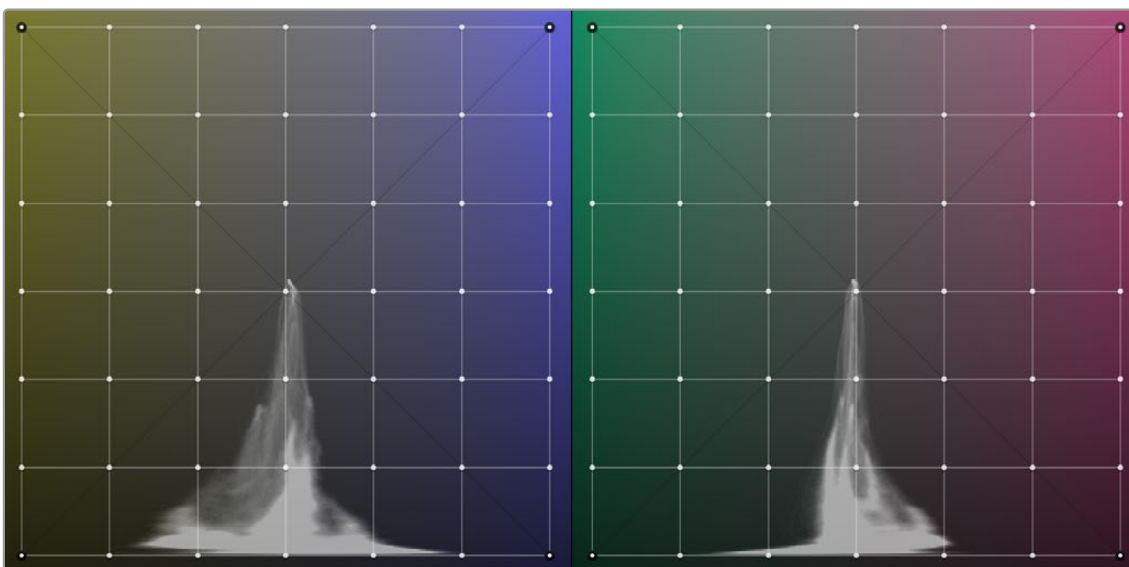
By default, the Hue-Saturation grid has a resolution of 6x6, which is the lowest resolution available. The Chroma-Luma grid defaults to 6x6, but can be set lower, to 4x4. Low resolution warping grids make it easy to create broad color adjustments affecting large ranges of analogous color with smooth results. Additionally, lower quality media that's 8-bit, with 4:2:0 chroma subsampling, and/or highly compressed benefits from lower resolution warping grids to avoid artifacts and keep color adjustments smooth.



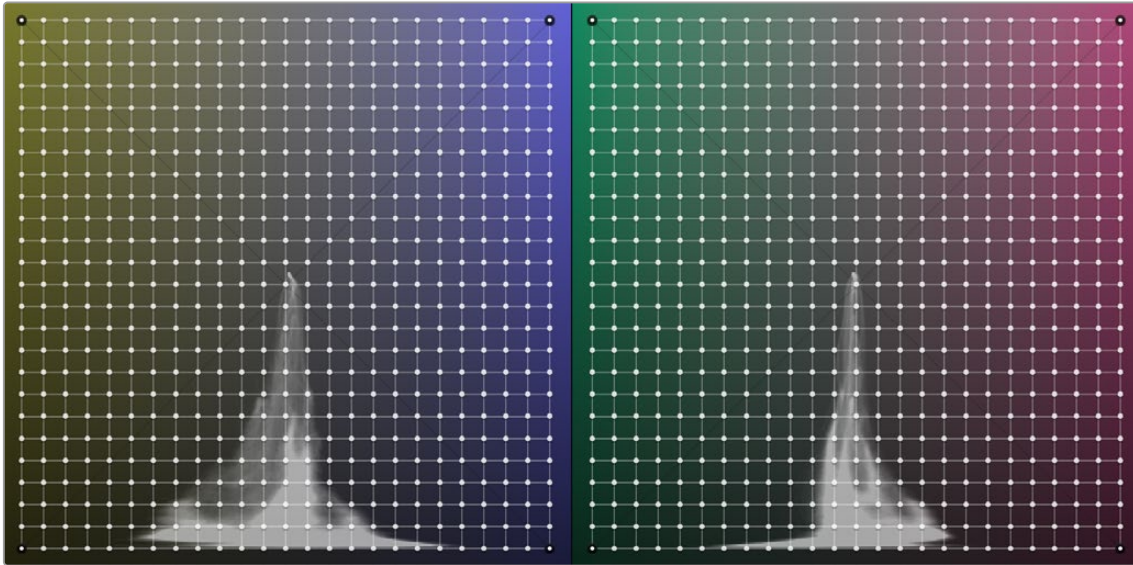
(Left) The Hue-Saturation grid set to a resolution of 6x6,
(Right) The Hue-Saturation grid set to a resolution of 24x16

On the other hand, higher resolution grids let you make more specific adjustments to tighter ranges of color. However, this approach is more useful in projects using high quality media (10- or 12-bit, 4:2:2 or 4:4:4 chroma subsampling, and minimally compressed) to avoid unwanted artifacts. If you're using high-quality media, you'll find you can use the Color Warper to make incredibly specific adjustments.

You can set the default Hue and Saturation grid resolutions in the Color Warper's Option menu by choosing Default Hue Resolution or Default Saturation Resolution and selecting your preferred resolution from the drop-down menu.



The Chroma-Luma grid set to a resolution of 6x6

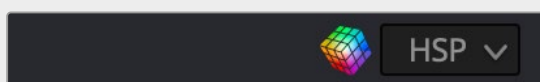


The Chroma-Luma grid set to a resolution of 24x24

NOTE: If you make adjustments with the grid at one resolution, it's possible to change the grid to a higher or lower resolution, and your adjustment will be interpolated to fit the new grid resolution, although the resulting adjustment may change a bit.

You Can Warp Color in Different Color Spaces

A drop-down menu at the bottom right of the warping grid lets you choose which color space to use for manipulating the colors of the image. Different color spaces project the colors of the image into the two-dimensional warping grid in different ways, and some may make it easier to manipulate a particular image in the way you want by spreading out different ranges of color more widely.



The color space found underneath the grid controls

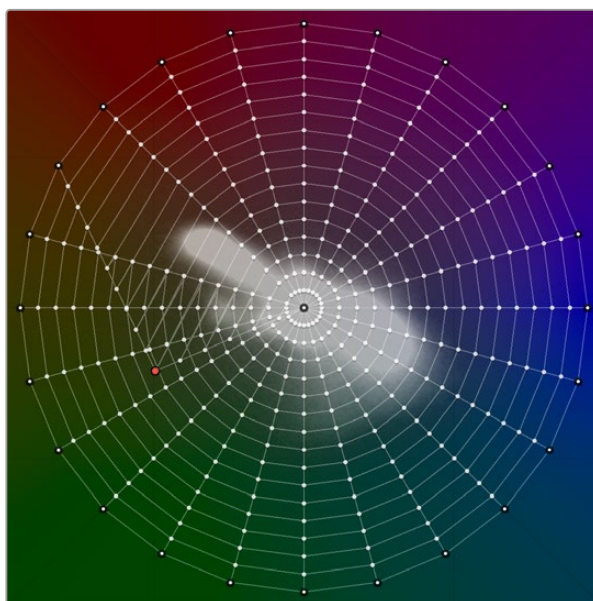
The available color spaces are:

- HSV
- HSL
- HSY
- HSP

Hue – Saturation Controls

As the name implies, the Hue-Saturation mode lets you alter hue and saturation simultaneously. For most people, this is likely the most intuitive mode in which to work as the range of hues and saturation in the image is represented radially, just like the Vectorscope; in fact a Vectorscope graph of the image appears underneath the “wheel” grid of control points used to manipulate color.

By default, dragging any point of this grid changes the colors immediately surrounding that point that are between the point being manipulated and adjacent “spokes” of the radial grid that either aren’t influenced by that point, or are locked into place. In fact, you can pull these radial control points well past the boundary of adjacent spokes of the grid to warp those colors to entirely different hues.



The selected control point being adjusted doesn’t affect points on adjacent columns of control points that stretch from the outer ring to the center of this control, even when the dragged point is moved all the way past.

When manipulating control points to alter color, adjustments are similar to pushing color around a color wheel. Moving a control point radially around the circumference of the grid changes the hue of the colors corresponding to that control point. Moving a control point closer to the center desaturates those colors, while moving a control point closer to the edge of the grid increases the saturation of those colors.

The following sections describe how the different tools available in this mode work to let you manipulate these radial control points in different ways.

Tools

Each of the available tools let you manipulate control points on the warping grid in different ways when you click on them.

- **Selection Tool:** The default tool mode. Selecting this tool lets you select control points either individually (by clicking on a single point or command-clicking multiple points) or collectively by dragging a bounding box over multiple points, and you can right-click on control points to de-select, unlock, and reset them. You can also Shift-click with this tool to lock points into place without moving them. This is the most useful tool to use when manipulating the image by clicking and dragging in the Viewer to sample a color and adjust it at the same time.

- **Draw Selection:** Selecting this tool lets you make a selection of control points by clicking and dragging to draw over all the points you want to select. This is good for making large, but specific, selections of points.
- **Pin/De-Pin:** Selecting this tool lets you pin multiple control points by either clicking on them or by clicking and dragging to draw over all the points you want to pin/de-pin.
- **Pull Points:** Selecting this tool lets you make adjustments by clicking anywhere on the warp grid, even between points, to pull all neighboring control points towards where you clicked. This can be used to reduce color contrast within a specific range of colors.
- **Push Points:** Selecting this tool lets you make adjustments by clicking anywhere on the warp grid, even between points, to push all neighboring control points within a particular proximity away from where you clicked. This can be used to increase color contrast within a specific range of colors.

Modifiers

Each of the Modifier buttons let you manipulate selections and selected control points on the warping grid in different ways immediately upon clicking each button.

- **Increase Falloff/Smooth Selection:** If you've selected one or more control points, clicking this button expands the selection to include all adjacent control points surrounding the selection.
- **Decrease Falloff/Smooth Selection:** If you've selected a group of control points, clicking this button shrinks the selection by deselecting the outermost ring of selected control points, leaving the inner control points selected.
- **Invert Selection:** Clicking this button selects all unselected control points, and de-selects all selected control points. This is useful when you want to make separate color adjustments to both specifically selected halves of the warping grid.
- **Convert Selection to Pin:** Clicking this button pins all currently selected control points.
- **Select/Pin Column:** If you have one or more control points selected, clicking this button expands the selection to include all points on every column that have at least one selected point.
- **Select/Pin Ring:** If you have one or more control points selected, clicking this button expands the selection to include all points on every ring around the center that have at least one selected point.
- **Select/Pin All, Deselect/Pin All:** Clicking this button toggles the selected state of the entire warping grid on or off.
- **Reset Selection/Pins:** If you have one or more control points selected, clicking this button resets their position to the original default position in the warping grid, without de-selecting them.

Range

The Range control is a fast method of selecting multiple control points corresponding to a specific range of colors.

- **Range:** A gradient shows the range of hues currently being presented in the warping grid. Dragging the left and right handles of the Range control selection box lets you automatically select all control points corresponding to the hues that appear within the selection box. This is a fast way of selecting all points within a range of colors for uniform manipulation.

Auto Lock Controls

The Auto Lock controls enables DaVinci Resolve to automatically lock a border of control points surrounding any control point you select and adjust, which makes highly-specific color adjustments easier to accomplish.

- **Auto Lock:** Enables and disables this behavior.
- **X Points Border:** Lets you set how many points away from the control point you're adjusting the border of locked control points that restricts your adjustments is. How large an area this ends up being depends on how many points you choose, and on the resolution of the warping grid. At higher grid resolutions, the same points distance isolates a smaller region of color.

Smoothing Controls

The smoothing controls let you “ease off” an adjustment you’ve made by progressively moving one or more selected control points towards their original default position in the warping grid.

- **Smooth Chroma:** Each click of this button rotates the angle of selected control points around the circumference of the circular warping grid towards their original position, bringing the hue of the adjusted colors closer and closer to the original hues of the image. Saturation is unaffected.
- **Reset Chroma:** Resets the angle of all selected control points to the original hues of those control points. Saturation adjustments are unaffected.
- **Smooth Saturation:** Each click of this button moves the position of a control point closer to its original position relative the center of the circular warping grid, thus bringing the saturation at that point closer to the original image saturation of the node's input. Hue is unaffected.
- **Reset Saturation:** Resets the distance from the center of all selected control points to the original image saturation at those control points.
- **Reset Luma:** Resets luma to the original image values.

Chroma – Luma Controls

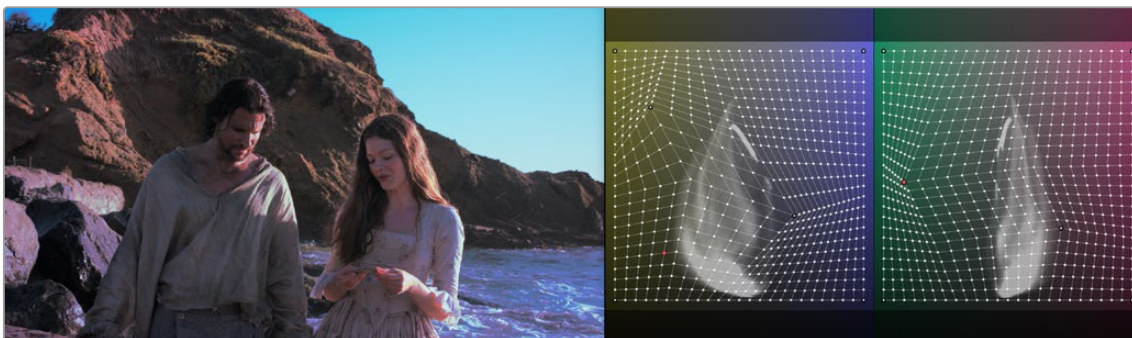
The Chroma-Luma mode lets you alter the hue and lightness of colors in the image simultaneously. This may not feel like a particularly intuitive way of working, as the grid controls are overlaid on colors projected as different sides of an RGB cube. However, this enables some powerful adjustments once you get the hang of how multiple adjustments interact in this mode, as well as the power of locking control points to limit your adjustments to specific areas of the two grids.

Whether you're sampling the image or manipulating the control points of this grid directly, dragging any point of this grid changes the color of the image corresponding to that point. Vertical adjustments change lightness, where up makes that part of the image lighter, and down makes that part of the image darker. Horizontal adjustments change hue, depending on which range of hues are shown in the two Chroma-Luma warping controls, and which of these you're adjusting.

By default, only the outer four corners of this control are locked, so any adjustment you make to any control point pushes and pulls all the other colors throughout the image, depending on your adjustment. Working this way, multiple adjustments to multiple colors gradually pins different colors into place, with each adjustment warping the colors falling in between to maintain a smooth transformation from one adjustment to the next. This can be a good way to make an overall stylistic adjustment to the image.



The original image



The result of making multiple interacting color adjustments across the entire Chroma Luma grid

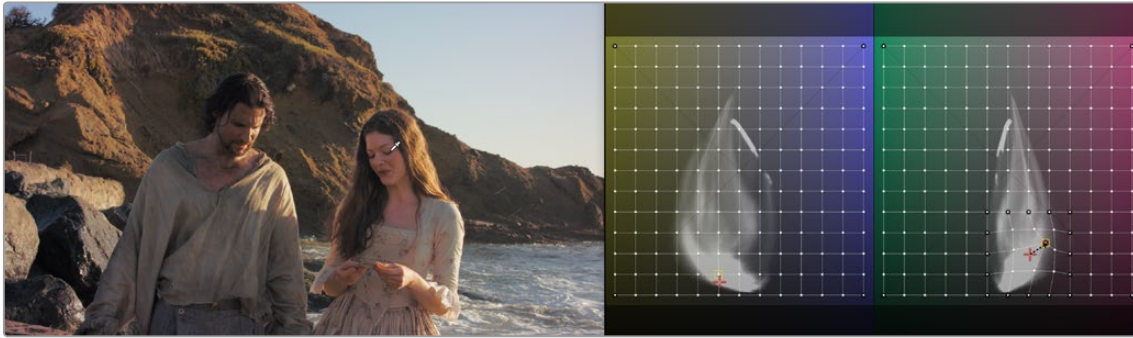
Another approach to using this mode is to make more targeted corrections by manually locking different colors in the image that you don't want to adjust. You can do this by using the Selection tool and Shift-clicking any value in the Viewer, or by Shift-clicking any control point on the grid control. By locking colors you don't want to change, you can focus on manipulating a more specific range of colors without altering the entire image.

Alternately, this is where the Auto Lock controls really shine. When you turn these on, you can choose the type of region you want to affect (vertical column, horizontal row, or a square region), then choose how large a region of color you want to manipulate (one or two points away from the point that's selected). Keep in mind that the current resolution of the warping grid also has an effect on how large the resulting region of color will be. Lower resolution grids will let you manipulate larger regions of color, while higher resolution grids will let you adjust a more narrow range of color.



The Auto Lock controls let you make highly targeted adjustments to specific sections of the warping grid.

Once these controls are enabled, simply clicking once on the image or warping grid selects a control point and automatically locks off the region of the grid you want to focus your adjustment on. In the following example, a medium-resolution grid is used in conjunction with the Auto Lock controls set to lock a 2-point rectangular region. So, clicking on the skin of the woman's face and dragging adjusts a section of available reds within the locked-off region of the grid.



Making a targeted color adjustment to a locked-off region of the Chroma Luma grid

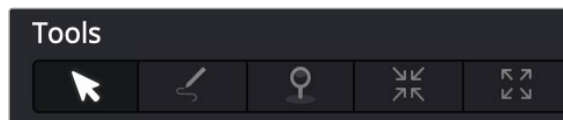
The following sections describe how the different tools available in this mode work to let you manipulate these radial control points in different ways.

Axis Angle

When you're in Chroma-Luma mode, an additional control appears underneath the warping grids, a slider named Axis Angle. Dragging this slider to the left or right changes the range of hues in each of the two warping grids that's presented, letting you manipulate different ranges of color.

Tools

Each of the available tools let you manipulate control points on the warping grid in different ways when you click on them.

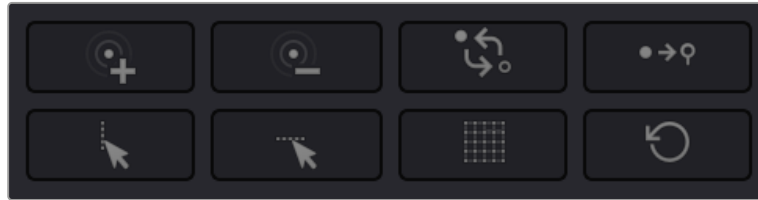


The Tools buttons

- **Grid 1/Grid 2 tabs:** Since the Chroma-Luma controls expose two grids representing two different ranges of hue and luma, these two tabs let you choose which grid will be manipulated when you sample the image in the Viewer, as well as which grid the Tools controls affect.
- **Selection Tool:** The default tool mode. Selecting this tool lets you select control points either individually (by clicking on a single point or command-clicking multiple points) or collectively by dragging a bounding box over multiple points. You can also Shift-click with this tool to lock points into place without moving them, and you can right-click on control points to de-select, unlock, and reset them. This is the most useful tool to use when manipulating the image by clicking and dragging in the Viewer to sample a color and adjust it at the same time.
- **Draw Selection:** Selecting this tool lets you make a selection of control points by clicking and dragging to draw over all the points you want to select. This is good for making large, but specific, selections of points.
- **Pin/De-Pin:** Selecting this tool lets you pin multiple control points by either clicking on them or by clicking and dragging to draw over all the points you want to pin/de-pin.
- **Pull Points:** Selecting this tool lets you make adjustments by clicking anywhere on the warp grid, even between points, to pull all neighboring control points towards where you clicked. This can be used to reduce color contrast within a specific range of colors.
- **Push Points:** Selecting this tool lets you make adjustments by clicking anywhere on the warp grid, even between points, to push all neighboring control points within a particular proximity away from where you clicked. This can be used to increase color contrast within a specific range of colors.

Modifiers

Each of the Modifier buttons let you manipulate selections and selected control points on the warping grid in different ways immediately upon clicking each button.



The Modifier buttons

- **Increase Falloff/Smooth Selection:** If you've selected one or more control points, clicking this button expands the selection to include all adjacent control points surrounding the selection.
- **Decrease Falloff/Smooth Selection:** If you've selected a group of control points, clicking this button shrinks the selection by deselecting the outermost ring of selected control points, leaving the inner control points selected.
- **Invert Selection:** Clicking this button selects all unselected control points, and de-selects all selected control points. This is useful when you want to make separate color adjustments to both specifically selected halves of the warping grid.
- **Convert Selection to Pin:** Clicking this button pins all currently selected control points.
- **Select/Pin Column:** If you have one or more control points selected, clicking this button expands the selection to include all points on every column that have at least one selected point.
- **Select/Pin Row:** If you have one or more control points selected, clicking this button expands the selection to include all points on every row that have at least one selected point.
- **Select/Pin All, Deselect/Pin All:** Clicking this button toggles the selected state of the entire warping grid on or off.
- **Reset Selection/Pins:** If you have one or more control points selected, clicking this button resets their position to the original default position in the warping grid, without de-selecting them.

Range

The Range control is a fast method of selecting multiple control points corresponding to a specific range of colors.

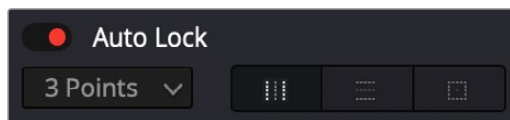


The Range control

- **Range:** A gradient shows the range of hues currently being presented in the warping grid. Dragging the left and right handles of the Range control selection box, or dragging within the range control to set both boundaries, lets you automatically select all control points corresponding to the hues that appear within the selection box. Once you've set a range, you can move the range to the left and right by dragging the center of the selection. This is a fast way of selecting all control points within a range of colors for uniform manipulation.

Auto Lock Controls

The Auto Lock controls enables DaVinci Resolve to automatically lock a border of control points surrounding any control point you select and adjust, which makes highly specific color adjustments easier to accomplish. These are particularly useful in the Chroma-Luma warping grid.

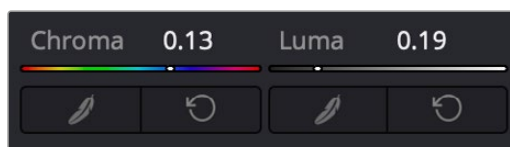


The Auto Lock controls

- **Auto Lock:** Enables and disables this behavior.
- **X Points Border:** Lets you set how many points away from the control point you're adjusting the border of locked control points that restricts your adjustments is. How large an area this ends up being depends on how many points you choose, and on the resolution of the warping grid. At higher grid resolutions, the same points distance isolates a smaller region of color.
- **Lock Column:** Lets you choose to restrict your adjustment to affect all control points within a particular column of the rectangular Chroma-Luma graph. The width of this column is defined by the border width controls. This is useful when you want to primarily adjust the lightness of a range of colors, while making a small adjustment to hues.
- **Lock Row:** Lets you choose to restrict your adjustment to affect all control points within a particular row of the rectangular Chroma-Luma graph. The height of this row is defined by the border width controls above. This is useful when you want to primarily adjust the hue of a range of colors, while making a small adjustment to lightness.
- **Lock Region:** Lets you choose to restrict your adjustment to affect all control points within a rectangular region surrounding the selected point. This is useful when you want to adjust a targeted region of color that corresponds to a well-defined feature of the image, such as a specifically colored hat, shirt, foliage, or skin tone. Smaller regions will make more specific adjustments than larger regions.

Smoothing Controls

If you've found that you've gone a bit too far with an adjustment and need to back off a little bit, the Smooth controls let you do so gradually, or start over if necessary.



The Smoothing controls

- **Smooth Chroma:** Each click of this button moves selected control points left or right, back towards their original position, bringing the hue of the adjusted colors closer and closer to the original hues of the image. Luma is unaffected.
- **Reset Chroma:** Resets the position of all selected control points to the original hues of those control points. Luma adjustments are unaffected.
- **Smooth Luma:** Each click of this button moves selected control points up or down, back towards their original position, bringing the luma of the adjusted colors closer and closer to the original luma of the image. Chroma is unaffected.
- **Reset Luma:** Resets the position of all selected control points to the original luma of those control points. Chroma adjustments are unaffected.